Trainings

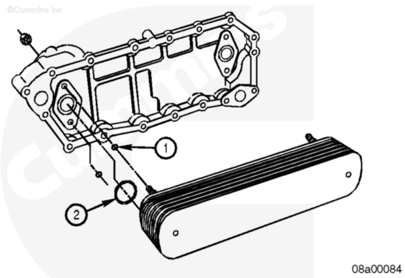
Preparatory Steps

- Remove the lubricating oil cooler. Refer to Procedure 007-003 (/qs3/pubsys2/xml/en/procedures/57/57-007-003.html).

Disassemble

Remove the flange nuts and the oil cooler element cover(s) from the oil cooler element(s).

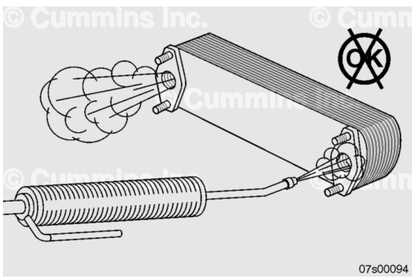
Remove and discard the oil cooler stud o-rings (1) and the oil passage o-rings (2).



Clean and Inspect for Reuse

Note : Cummins, Inc. does not recommend any form of cleaning of oil cooler elements (internal or external). This includes cleaning with solvent, flushing with oil, steam cleaning, and ultrasonic cleaning.

Note : Replace the elements if any debris is found or if the engine has had a debris-causing malfunction.



If the engine has experienced a malfunction that allows debris to pass through the lubrication system to the full-flow oil filters and oil passages resulting in the lubricating oil cooler to become contaminated, the oil cooler elements **must** be replaced.

Inspect the oil cooler elements externally for scale build-up, corrosion, embedded debris, and/or any damage to the elements. If there is any evidence of any of these conditions, the oil elements **must** be replaced.

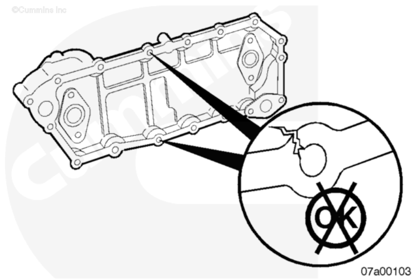
Note : Contamination can be defined by metal, rubber, or other solid particles present in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of any other components such as they do not meet reuse guidelines.

Remove gasket material from the oil cooler cover, oil cooler element, and the engine block mating surface.

Use solvent to clean the oil cooler housing and cover.

Check the cover and housing for cracks. If leaks are suspected, use the dye-penetrant method to locate them.

Check the cover and housing for corrosion.

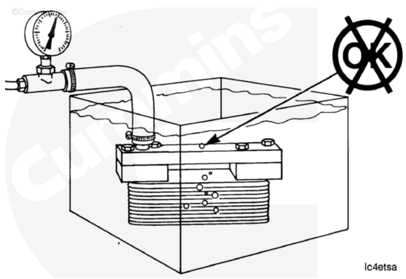


Pressure Test

Pressure test the cores to check for leaks.

Measurements		
	kpa	psi
Air Pressure:	415	60

Note : Heating the water in the tank to 50°C [122°F] will improve the test results.



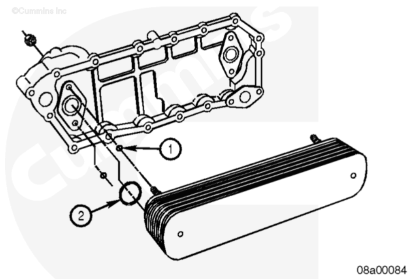
Assemble

Install the o-rings onto the oil cooler element studs (1).

Install the oil cooler elements into the oil cooler housings.

Install the flange nuts onto the studs.

Tighten the flange nuts.



Torque Value:
10-mm Studs 34 n•m [25 ft-lb]

Torque Value:
12-mm Studs 65 n•m [48 ft-lb]

Note : Verify all packing tape has been removed before assembly onto the engine.

Finishing Steps

- Install the oil cooler. Refer to Procedure 007-003 (/qs3/pubsys2/xml/en/procedures/57/57-007-003.html).
- Operate the engine and check for leaks.

Last Modified: 09-Jun-2016